A Survey on Foundation Language Models for Single-cell Biology

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Abstract

The recent advancements in language models have significantly catalyzed progress in computational biology. A growing body of research strives to construct unified foundation models for single-cell biology, with language models serving as the cornerstone. In this paper, we systematically review the developments in foundation language models designed specifically for single-cell biology. Our survey offers a thorough analysis of various incarnations of single-cell foundation language models, viewed through the lens of both pre-trained language models (PLMs) and large language models (LLMs). This includes an exploration of data tokenization strategies, pre-training/tuning paradigms, and downstream single-cell data analysis tasks. Additionally, we discuss the current challenges faced by these pioneering works and speculate on future research directions. Overall, this survey provides a comprehensive overview of the existing single-cell foundation language models, paving the way for future research endeavors.

1 Introduction

In recent years, the rise of language models (Vaswani, 2017; Devlin, 2018; Radford et al., 2018; Raffel et al., 2020; Zhu et al., 2025) has driven the flourishing development of research in various fields. Among them, the intersection of computational biology and language models holds great promise and has gained increasing attention. This innovative research direction enables researchers to explore the interpretation of cells as languages (Theodoris et al., 2023; Levine et al., 2023) and leverage language models as a cornerstone to construct foundation models. These foundation language models have demonstrated their ability to obtain robust and generalizable cell representations across various datasets and tasks. Consequently, they have shown remarkable performance in a variety of single-cell data analysis tasks, surpassing the capability of specialist models (Zhang et al., 2024), and promoting developments in various healthcare domains (Wu et al., 2024), including disease diagnosis, drug discovery, and immunotherapy.

Existing single-cell foundation language models can be broadly divided into two groups: single-cell pre-trained language models (PLMs) and single-cell large language models (LLMs).[1] In studies centered on single-cell PLMs (Yang et al., 2022; Cui et al., 2024), genes are typically treated as tokens in most cases, allowing cells to be represented as sentences. Researchers can then leverage well-established language models in the field of natural language processing (NLP), such as BERT (Devlin, 2018) and GPT (Radford et al., 2018), to perform large-scale pre-training from scratch on cells. This process strives to derive unified cell representations, which can be applied for zero-shot inference or fine-tuning across various single-cell data analysis tasks. In contrast, research based on single-cell LLMs (Levine et al., 2023; Chen and Zou, 2024) often does not require pre-training on extensive cells. Instead, these investigations leverage powerful LLMs with proven emergent capabilities. By converting cells into textual formats, LLMs can interpret cells and can be effectively utilized for various single-cell data analysis tasks after tuning.

To assist researchers in organizing their thoughts and shed light on future research endeavors, we systematically review the existing literature on single-cell foundation language models. By conducting a thorough analysis of data tokenization strategies, pre-training or fine-tuning paradigms, and their applications in a wide spectrum of single-cell data analysis tasks, we meticulously examine and evalu-

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[1] In this paper, "single-cell PLMs" denotes cell-only models that pre-trained on single-cell data, while "single-cell LLMs" refers to text-only models that have not undergone pre-training specifically on single-cell data.

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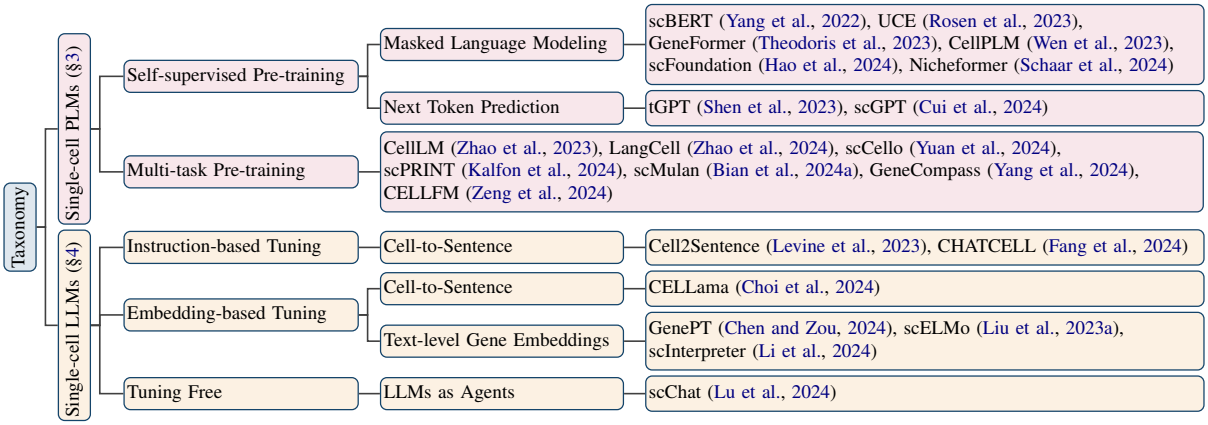


Figure 1: Taxonomy of foundation language models for single-cell biology.

ate each of the current models. Furthermore, drawing from this prior analysis, we illuminate the array of challenges encountered in the pursuit of developing unified single-cell foundation language models. Given these challenges, we subsequently propose potential pathways for future research that aim to integrate language models with single-cell biology.

While several surveys have been conducted on Transformer-based single-cell models (Lan et al., 2024; Szałata et al., 2024; Bian et al., 2024b), they have not yet provided a comprehensive exposition and analysis from the perspective of language modeling. As shown in Figure 1 and Figure 2, we provide a clear taxonomy and workflow of single-cell foundation language models to facilitate understanding. Overall, the main contributions of this paper can be summarized as follows:

- ***First Survey.*** To the best of our knowledge, we are the first to present a comprehensive survey about foundation language models for single-cell biology, thoroughly analyzing the techniques and applications of existing models.
- ***Novel Perspectives.*** Based on the approaches in interpreting cells, we categorize existing models into single-cell PLMs and LLMs, and then conduct a comprehensive analysis of them.
- ***New Frontiers.*** We discuss and highlight the challenges in constructing unified foundation language models for single-cell biology, which pave the way for future research endeavors.

2 Preliminaries

We start by providing definitions and mathematical notations for single-cell foundation language models. Unless stated otherwise, all formulations in this paper follow these definitions and notations.

Encoding. Consider a cell-by-gene matrix $X \in \mathbb{R}^{N \times G}$, where N and G denote the number of cells and genes collected in a specific dataset, respectively. In this format, cell data exhibits a structure similar to text data, allowing the use of language models for modeling cell representations. In most studies, each gene is treated as a token, enabling cells to be understood as sentences by language models. Let $\mathcal{LM}$ denote a language model, then the encoding process can be formulated as:

$$z = \mathcal{LM}(Tokenizer(x)), \tag{1}$$

where $x \in \mathbb{R}^{1 \times G}$ is a cell sample drawn from X, and $z \in \mathbb{R}^{1 \times D}$ denotes the corresponding D-dimensional cell representation.

Pre-training. In the pre-training paradigm, cell representation z is fed into a pre-training head $\mathcal{PH}$ to obtain specialized feature h for pre-training using the learning objective $\mathcal{L}_{\mathcal{P}}$:

$$h = \mathcal{PH}(z), \tag{2}$$

$$\mathcal{L}_{\mathcal{P}} = PT(h, y^p), \tag{3}$$

where PT and y^p represent different pretext tasks for pre-training and their corresponding labels. Based on the specific pretext task, the label can be a real class label (supervised pre-training), a self-constructed label (self-supervised pre-training), or a hybrid label (multi-task pre-training).

Tuning. Similarly, in the tuning paradigm, cell representation z is sent into a task-specific head $\mathcal{TH}$ to obtain prediction p for tuning with the learning objective $\mathcal{L}_{\mathcal{T}}$:

$$p = \mathcal{TH}(z), \tag{4}$$

$$\mathcal{L}_{\mathcal{T}} = DT(p, y^t), \tag{5}$$

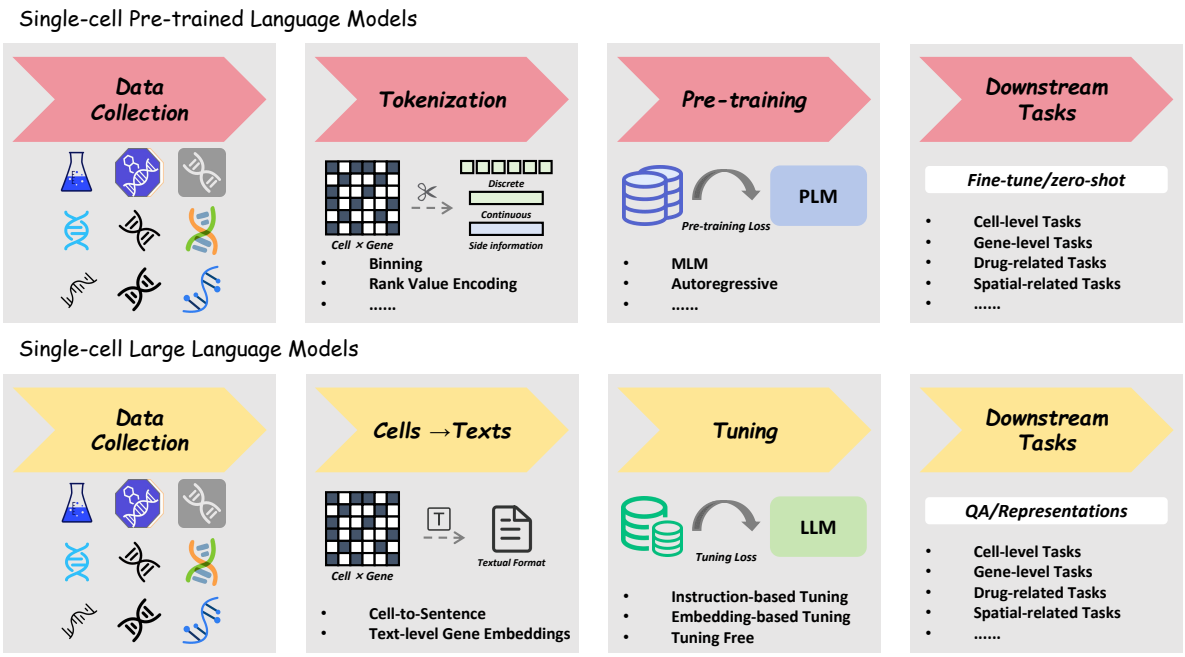


Figure 2: Workflow of PLMs and LLMs for single-cell biology.

where DT and y^t denote various downstream tasks and the corresponding ground truth labels. Depending on the specific scenario, pre-training and tuning tasks can be either consistent or inconsistent.

3 PLMs for Single-cell Biology

In this section, we provide a holistic review of existing PLMs for single-cell biology, including their data tokenization strategies (Section 3.1) and pre-training paradigms (Section 3.2). A summary of these single-cell PLMs is shown in Table 1.

3.1 Tokenization Strategies

To encode cells using language models, the first obstacle is to transform cells into a format that language models can understand. On the one hand, the process of tokenizing cells is challenging due to the lengthy nature and the inherent sparsity of gene sequences. On the other hand, since gene sequences are unordered, it is often difficult to unify cell data from different sources, posing difficulties for large-scale pre-training. To address this issue, several approaches have proposed various data tokenization strategies from the following perspectives.

Discrete Tokens. In the field of NLP, textual format data needs to be transformed into discrete tokens before it can be understood by a language model (Harris, 1954; Mikolov, 2013; Rong, 2014). For single-cell transcriptomics data composed of gene sequences in continuous form, discretization is also an important step in tokenization. For example, scBERT (Yang et al., 2022) and CellLM (Zhao et al., 2023) utilize a straightforward binning technique that converts log-transformed continuous gene expression values into the nearest integer discrete values, while also setting a ceiling value. Another approach to discretizing cells is rank value encoding, which is proposed by GeneFormer (Theodoris et al., 2023) and widely used in the GeneFormer family (GeneFormer (Theodoris et al., 2023), LangCell (Zhao et al., 2024), and scCello (Yuan et al., 2024)). This strategy involves sorting the frequencies of gene names and encoding them using a gene vocabulary. The maximum length of each cell is set to 2048, and any remaining positions are padded with the *<PAD>* symbol.

Continuous Embeddings. A more popular strategy uses continuous embeddings for fine-grained information. For example, UCE (Rosen et al., 2023) and scPRINT (Kalfon et al., 2024) utilize large protein language models to obtain continuous embeddings for each gene, which are then used to tokenize cells. CellPLM (Wen et al., 2023) multiplies each gene in the normalized cell data with a randomly initialized learnable layer to obtain continuous cell embeddings. scFoundation (Hao et al., 2024) employs a hierarchical Bayesian downsampling strategy to maintain the quality and consistency of large-scale data. CellFM (Zeng

Model	Tokenization Strategy	Pre-training Paradigm	Pre-training Data
scBERT (Yang et al., 2022)	Binning	Masked Language Modeling	1M scRNA-seq Data
UCE (Rosen et al., 2023)	Protein-coding	Masked Language Modeling	36M Cells
GeneFormer (Theodoris et al., 2023)	Rank Value Encoding	Masked Language Modeling	27.4M Transcriptomics Data
CellPLM (Wen et al., 2023)	Cells as Tokens	Masked Language Modeling	11M Cells
scFoundation (Hao et al., 2024)	Downsampling	Masked Language Modeling	50M scRNA-seq Data
Nicheformer (Schaar et al., 2024)	Ranking, Metadata	Masked Language Modeling	57M Cells
tGPT (Shen et al., 2023)	Ranking, Padding	Next Token Prediction	22.3M Cells
scGPT (Cui et al., 2024)	Binning, Metadata	Next Token Prediction	33M Cells
CellLM (Zhao et al., 2023)	Binning	Multi-task Pre-training	2M scRNA-seq Data
LangCell (Zhao et al., 2024)	Rank Value Encoding	Multi-task Pre-training	27.5M Cells and Texts
scCello (Yuan et al., 2024)	Rank Value Encoding	Multi-task Pre-training	22M Cells
scPRINT (Kalfon et al., 2024)	Protein-coding	Multi-task Pre-training	50M Cells
scMulan (Bian et al., 2024a)	Cell Sentence	Multi-task Pre-training	10M Transcriptomics Data
GeneCompass (Yang et al., 2024)	Ranking, Metadata	Multi-task Pre-training	126M Transcriptomics Data
CellFM (Zeng et al., 2024)	Padding, MLP	Multi-task Pre-training	100M Cells

Table 1: Summary of pre-trained language models for single-cell biology.

et al., 2024) uses padding to align large-scale data and then applies a multilayer perceptron (MLP) for projection. Nicheformer (Schaar et al., 2024) and GeneCompass (Yang et al., 2024) combine ranked gene information with metadata for continuous value embeddings. tGPT (Shen et al., 2023) adopts padding and ranked gene information for direct continuous encoding of large-scale cell data. scGPT (Cui et al., 2024) incorporates metadata information after binning to transform the embeddings from discrete to continuous values. scMulan (Bian et al., 2024a) combines the original data, metadata, and task placeholders to form cell sentences for obtaining continuous cell embeddings.

Side Information. In addition to the raw cell data, many approaches incorporate side information during the tokenization process. One mainstream approach (Nicheformer (Schaar et al., 2024), scGPT (Cui et al., 2024), scMulan (Bian et al., 2024a), and GeneCompass (Yang et al., 2024)) is to utilize metadata to obtain side information. The available metadata attributes include cell state, cell type, organ source and specific region, donor age and gender, and sequencing technology. Another approach is to leverage existing biological foundation models as bridges. For instance, UCE (Rosen et al., 2023) and scPRINT (Kalfon et al., 2024) obtain the side information from large protein language models, which bridge the gap between cells and languages through protein information.

3.2 Pre-training Paradigms

An important step in building single-cell foundation language models from scratch is large-scale pre-training. After tokenizing cell data into a format understandable by language models, we can conduct pre-training on a vast volume of cell data using similar pretext tasks in NLP. Next, we will elaborate on these pre-training tasks.

Masked Language Modeling. Numerous studies (Salazar et al., 2019; Min et al., 2023; Minaee et al., 2024) have demonstrated that models can learn high-quality representations with generalization capabilities through reconstructing the masked items (e.g., tokens, patches, and features), laying the foundation for self-supervised learning. The remarkable success of masked language modeling (MLM) in the domains of language (Devlin, 2018; Liu, 2019) and vision (He et al., 2022; Bao et al., 2021) also inspires its application in single-cell biology. For single-cell PLMs, the mainstream masking strategies involve randomly masking genes with a certain probability (e.g., 15%-30%) (Yang et al., 2022; Rosen et al., 2023; Theodoris et al., 2023; Hao et al., 2024; Schaar et al., 2024), or using the Gaussian mixture distribution as priors for masking measured genes (Wen et al., 2023).

Next Token Prediction. Although autoregressively predicting the next token has dominated the pre-training paradigms of LLMs (Touvron et al., 2023; Bai et al., 2023a; Achiam et al., 2023) and multi-modal foundation models (Bai et al., 2023b; Wang et al., 2024), this approach has not yet become popular for single-cell PLMs. Only tGPT (Shen et al., 2023) and scGPT (Cui et al., 2024) adopt this strategy for pre-training language models from scratch. We attribute this phenomenon to two reasons: (1) Compared to the vast amount

Model	Base Model	From Cells to Texts	Tuning Paradigm
Cell2Sentence (Levine et al., 2023)	GPT-2	Cell-to-Sentence	Instruction-based
CHATCELL (Fang et al., 2024)	T5	Cell-to-Sentence	Instruction-based
scInterpreter (Li et al., 2024)	LLaMA-13B	Text-level Gene Embeddings	Embedding-based
GenePT (Chen and Zou, 2024)	GPT-3.5	Text-level Gene Embeddings	Embedding-based
scELMo (Liu et al., 2023a)	GPT-3.5	Text-level Gene Embeddings	Embedding-based
CELLama (Choi et al., 2024)	all-MiniLM-L12-v2	Cell-to-Sentence	Embedding-based
scChat (Lu et al., 2024)	GPT-4o	None	Tuning Free

Table 2: Summary of large language models for single-cell biology.

of text data, existing single-cell datasets are still relatively small, making the autoregressive training process insufficient for strong generalization ability. (2) Cell data naturally exhibits sparsity, resulting in most ground truth values for predicting the next token being zero. This can lead the model to learn trivial solutions during the training process.

Multi-task Pre-training. In contrast to a single pretext task, many approaches opt for multi-task pre-training, including both self-supervised and supervised tasks. Among them, MLM remains the most fundamental task. In addition to MLM, cell generation (Bian et al., 2024a) and contrastive learning (Zhao et al., 2023, 2024; Yuan et al., 2024; Kalfon et al., 2024) are also widely employed. Overall, multi-task pre-training incorporates supervised tasks using labels or other side information on top of self-supervised tasks. These supervised tasks include classification (Zhao et al., 2023; Kalfon et al., 2024; Bian et al., 2024a; Yang et al., 2024; Zeng et al., 2024), cell-text matching (Zhao et al., 2024), bottleneck learning (Kalfon et al., 2024), and metadata prediction (Bian et al., 2024a). Furthermore, some approaches employ denoising (Kalfon et al., 2024) and cell ontology alignment (Yuan et al., 2024) to capture gene-gene and cell-cell structural information, respectively.

4 LLMs for Single-cell Biology

Due to the high cost required for pre-training language models from scratch, another line of research chooses to build upon powerful LLMs and facilitate their understanding of single-cell biology. In this section, we will specifically introduce single-cell LLMs, including techniques to convert single-cell data into textual formats (Section 4.1) and tuning paradigms (Section 4.2). We also provide a summary of existing single-cell LLMs in Table 2.

4.1 From Cells to Texts

To boost the reasoning abilities of LLMs for single-cell data analysis tasks, the first step is to transform single-cell transcriptomics data into textual formats. Currently, the existing transformation approaches can be divided into the following two groups.

Cell-to-Sentence. Cell2Sentence (Levine et al., 2023) proposes to use textual gene names to represent cells. For a set of single-cell ribonucleic acid sequencing (scRNA-seq) data, genes are sorted according to their expression values after normalization. Subsequently, for each cell, the top 100 genes with the highest expression values are selected to form a cell sentence composed of gene names. This transformation approach not only resolves the issue of disorderliness in cell data but also addresses the variability in gene names across different datasets, which is also adopted by CHATCELL (Fang et al., 2024) and CELLama (Choi et al., 2024).

Text-level Gene Embeddings. Another transformation approach (Chen and Zou, 2024; Liu et al., 2023a; Li et al., 2024) involves prompting LLMs to inquire about the function of each gene and encoding their responses into text-level embeddings. Subsequently, the gene expression values at each position of a specific cell sample are used as weights to combine the text-level gene embeddings, through either ranking or weighted averaging methods.

4.2 Tuning Paradigms

Even though the construction of single-cell LLMs does not necessitate extensive pre-training like PLMs, several tuning approaches have been proposed to align LLMs with domain knowledge in single-cell biology. In this section, we will specifically introduce the existing tuning paradigms.

Instruction-based Tuning. Instruction-based tuning (Longpre et al., 2023; Zhang et al., 2023; Peng et al., 2023; Liu et al., 2024) is a prevalent

paradigm when fine-tuning LLMs. It involves transforming specific tasks and collecting data pairs in <*question, answer*> formats to fine-tune language models. In single-cell biology, this concept was first introduced by Cell2Sentence (Levine et al., 2023) and further developed by CHATCELL (Fang et al., 2024). Taking cell type annotation as an example, the task and cells can be transformed into questions by prompt templates, with cell types serving as the corresponding answers. This approach enables the collection of paired data formats for instruction-based tuning. However, instruction-based tuning faces notable limitations, as many single-cell data analysis tasks cannot be easily transformed into <*question, answer*> formats.

Embedding-based Tuning. Due to the limitations of instruction-based tuning, embedding-based tuning remains the mainstream paradigm. For methods based on text-level gene embeddings (Chen and Zou, 2024; Liu et al., 2023a; Li et al., 2024), the embeddings corresponding to each cell or gene can be directly obtained. For methods based on cell-to-sentence (Choi et al., 2024), the features extracted by sentence Transformer models (Reimers and Gurevych, 2019, 2020; Thakur et al., 2021) can be treated as embeddings. Then these embeddings and the corresponding ground truth labels are used for supervised fine-tuning.

Tuning Free. In addition to the above two tuning paradigms, there is another tuning free paradigm that treats LLMs as agents (Lu et al., 2024). This approach enables LLMs to be programmed to generate Python code for a wide range of single-cell data analysis tasks with raw data as input.

5 Single-cell Data Analysis Tasks

In this section, we delve into various single-cell data analysis tasks linked to single-cell foundational language models. Predominantly, these tasks encompass cell-level and gene-level analyses. Furthermore, there also exist models tailored towards drug-related, spatial-related, and other tasks.

5.1 Cell-level Tasks

As depicted in Appendix Table 3, mainstream cell-level tasks include cell type annotation (Cao et al., 2020b; Pasquini et al., 2021), discovery of novel cell types (Aevermann et al., 2018; Saviano et al., 2020), batch effect correction (Zhou et al., 2019; Tran et al., 2020), cell clustering (Kiselev et al., 2017, 2019), multi-omics data integration (Subramanian et al., 2020; Kang et al., 2022), and cell generation (Bergmann et al., 2015; Luo et al., 2024). Among these tasks, the most fundamental one is cell type annotation. Similar to traditional deep learning-based models (Shao et al., 2021; Chen et al., 2023), language models typically utilize a classification head after obtaining universal cell representations to make predictions on cell types, followed by fine-tuning or zero-shot inference on downstream datasets. Additionally, some models employ similarity comparison methods for annotation (Zhao et al., 2024) or transform the annotation task into a question-answering task (Fang et al., 2024). Building upon annotation, some models further extend their recognition capabilities to the discovery of novel cell types. These approaches often rely on assessing similarities between base cell types and novel cell types (Zhao et al., 2024) or the confidence of predictions (Yang et al., 2022). In scenarios where cell type information is lacking, many models typically resort to cell clustering to assess the discriminative nature of cell representations. Given that cell data is often collected across different laboratory settings (Leek et al., 2010; Lazar et al., 2013; Goh et al., 2017), it exhibits a pronounced batch effect. Consequently, many models pool data from different batches and then evaluate the effectiveness of batch effect correction through clustering performance. Since most language models are pre-trained on scRNA-seq data, they rarely focus on multi-omics tasks. Only scGPT (Cui et al., 2024) and scELMo (Liu et al., 2023a) have validated their ability to integrate multi-omics data, such as RNA and protein. In addition to the aforementioned discriminative tasks, generative tasks are also employed to assess the performance of language models, as they align closely with the pre-training objectives. Among these, prevalent tasks include conditional cell generation (Bian et al., 2024a) and unconditional cell generation (Levine et al., 2023; Fang et al., 2024).

5.2 Gene-level Tasks

We also present common gene-level tasks in Appendix Table 3, which could be broadly divided into three types: gene network analysis (Zhang and Horvath, 2005; van der Zwaag et al., 2009), gene perturbation (Lee et al., 2008; Meinshausen et al., 2016), and gene prediction (Mathé et al., 2002; Maji and Garg, 2013). Gene network analysis aims to elucidate cell functionality by examining inter-

actions between genes, encompassing uncovering gene networks (Cui et al., 2024), inferring gene regulation networks (Hao et al., 2024; Yang et al., 2024; Zeng et al., 2024), exploring gene programs (Chen and Zou, 2024), and conducting in silico treatment analysis (Theodoris et al., 2023; Liu et al., 2023a). The task of gene perturbation refers to perturbing specific genes, such as through knockouts, and observing how these changes affect cells or other genes. This process helps us understand gene functionality and the interactions between genes. Another task to directly understand genes is gene prediction, encompassing marker gene prediction (Yuan et al., 2024), gene expression and sensitivity prediction (Yang et al., 2024), as well as gene function, property, and interaction prediction (Levine et al., 2023; Chen and Zou, 2024).

5.3 Drug-related Tasks

Drug-related tasks (Appendix Table 4) (Tatonetti et al., 2012; Costello et al., 2014; Adam et al., 2020) aims to comprehend the mechanisms of drug action within biological systems, facilitating the selection and optimization of therapeutic regimens. The specific tasks in this field can be categorized into drug sensitivity prediction (Zhao et al., 2023; Fang et al., 2024) and drug response prediction (Hao et al., 2024; Yuan et al., 2024; Yang et al., 2024). The former focuses on forecasting the sensitivity of cells to specific drugs, while the latter primarily predicts the cellular response to drug treatment.

5.4 Spatial-related Tasks

Single-cell spatial transcriptomics (van den Brink et al., 2020; Weber, 2021; Longo et al., 2021; Piwecka et al., 2023) combines the advantages of single-cell transcriptomics and spatial transcriptomics, providing spatial context information of cells. This approach offers crucial insights into how cells interact with each other and their environment, and how these interactions influence cellular function and behavior. It is a powerful tool for studying complex tissues and diseases such as cancer or neurological disorders. As a result, some language models incorporate spatial information and specifically focus on spatial-related tasks (Appendix Table 4), including spatial transcriptomic imputation (Wen et al., 2023), spatial label prediction and composition (Schaar et al., 2024), as well as spatial context analysis (Choi et al., 2024).

5.5 Other Relevant Tasks

In addition to the aforementioned tasks, several single-cell foundation language models have been evaluated with other specific tasks (Appendix Table 4), including scRNA-seq denoising (Wen et al., 2023), inference of developmental lineages (Shen et al., 2023), cell-text retrieval (Zhao et al., 2024), pathway identification (Zhao et al., 2024), cell label prediction (Levine et al., 2023), and in-depth analysis of response from LLMs (Lu et al., 2024).

6 Challenges and Future Directions

Despite the advancements in current single-cell foundation language models, there are still several substantial challenges that warrant attention. It is crucial to understand these challenges as it will pave the way for future research endeavors.

6.1 Data Quality

Due to the following reasons, data quality has consistently been a hindrance to the development of unified single-cell foundation language models.

Inherent Sparsity. Single-cell data typically exhibits sparse characteristics (Aparicio et al., 2020; Baruzzo et al., 2020; Park and Lee, 2024), in that gene sequences are frequently lengthy, yet only a minor fraction (less than 10%) of genes exhibit measurable expression levels. This presents a significant gap compared to text data, as the information density in cells is much lower. Consequently, this poses challenges for language models in effectively understanding and interpreting cells.

Positional Information. Another characteristic of single-cell data is the irrelevance of gene order (Cao et al., 2020a; Xu et al., 2024). For a given cell, the arrangement of genes lacks a standardized convention and is only dependent on the preference of the sequencer. As a result, a cell can have multiple possible gene orderings, posing a challenge in constructing unified cell representations. Existing language models typically employ Transformer architectures (Vaswani, 2017; Dosovitskiy, 2020), which can easily learn positional and sequential information due to the presence of positional encoding. However, this often introduces noise when attempting to build unified cell representations.

Batch Effect. Pre-training requires collecting a large amount of single-cell data from multiple sources. However, due to the inconsistency in sequencing technologies (Grün and van Oudenaarden,

2015; Kashima et al., 2020), platforms (Zare and Kim, 2010; Valihrach et al., 2018), and laboratory environments (Carlo and Lee, 2006; Kalisky et al., 2011), different batches of single-cell data often exhibit severe batch effects. For the same type of cells from different batches, models tend to provide significantly different predictions, posing a challenge in obtaining unified cell representations.

Omics Resource. Existing datasets predominantly consist of scRNA-seq transcriptomics, while high-quality datasets for other omics, such as deoxyribonucleic acid (DNA) (Karemaker and Vermeulen, 2018; Evrony et al., 2021) and protein (Wu and Singh, 2012; Nasseri et al., 2011), are relatively scarce. This limits the ability of language models to analyze and process single-cell multi-omics data.

Direction. Considering the challenges above, it is crucial to address them and improve the quality of single-cell data for universal cell representations.

6.2 Model Design

There is still room for improvement in the structure of existing single-cell foundation language models, primarily due to the absence of a unified cell tokenizer and the emergence of scaling laws.

Cell Tokenizer. Although previous works have attempted to establish gene vocabularies (Theodoris et al., 2023; Cui et al., 2024) and transform all genes into vectors using word2vec techniques (Grohe, 2020; Rong, 2014) borrowed from NLP, we still have a long way to go toward a unified cell tokenizer. On the one hand, the largest gene vocabulary currently contains only around 70,000 genes, which is insufficient to encompass all the genes discovered in humans and mice (Bedell et al., 1997; Ingersoll et al., 2010; Bouabe and Okkenhaug, 2013). On the other hand, the integration of newly discovered genes into existing cell tokenizers remains unexplored.

Scaling Law. The development of foundation language models for single-cell biology significantly lags behind that of other domains (Cherti et al., 2023; Bahri et al., 2024). This is primarily due to limitations in data and model design. Currently, the largest single-cell PLMs have less than 1B parameters, which is relatively small compared to NLP models. Alternatively, single-cell LLMs heavily rely on textual embeddings to construct cell representations and necessitate fine-tuning on specific datasets. These factors indicate that scaling laws have not yet emerged in this domain.

Direction. Taking the above challenges into account, a potential future research direction could involve constructing a unified and scalable cell tokenizer, while also scaling up the model size to facilitate the emergence of intelligent capabilities.

6.3 Evaluation Protocol

The evaluation protocol for single-cell foundation language models lacks openness and transparency, primarily due to a scarcity of benchmarks and existing gaps between different domains.

Benchmark. Most of the existing single-cell foundation language models are validated on private datasets and different tasks, which can result in unfair comparisons between models. Currently, there is no publicly available benchmark with standardized datasets and evaluation metrics to assess the performance of these models in an open and fair manner. This hinders our understanding of the strengths and weaknesses of each model.

User-friendliness. The evaluation of single-cell foundation language models demands a substantial understanding of genomics and cell biology, making it less accessible for users without a background in biology. A major challenge lies in how to leverage insights from other domains (Huang et al., 2023; Tang et al., 2024) to transform various tasks into user-friendly question-answer formats.

Direction. Therefore, there is great anticipation for progress in establishing benchmarks to evaluate single-cell foundation language models and transforming various tasks into user-friendly formats.

7 Conclusion

This paper presented a systematic and comprehensive survey of foundation language models for single-cell biology. We categorized the existing models into single-cell PLMs and LLMs. Then we conducted a thorough literature review on data tokenization and pre-training paradigms for PLMs, as well as the techniques for converting cells into texts and tuning paradigms for LLMs. Additionally, we introduced various types of single-cell data analysis tasks to evaluate language models, including cell-level, gene-level, drug-related, spatial-related, and other tasks. Finally, we analyzed the challenges and future research directions from three perspectives: data quality, model design, and evaluation

protocol. Overall, this survey provides the first comprehensive analysis and summary of single-cell foundation language models, aiming to inspire researchers and encourage greater participation in this field, ultimately contributing to the development of universal foundation models.

Limitations

This survey provides a comprehensive review of single-cell foundation language models. However, there are still some limitations. Firstly, despite our efforts to cover as many relevant works published before the submission date, there is a possibility that a small portion of the literature may have been overlooked. Additionally, we primarily focus on the technical aspects of each model and do not extensively analyze the biological motivations and significance behind their design. We encourage readers to refer to the original papers for a more in-depth understanding in this regard. Moreover, investigating how single-cell foundation language models specifically enhance downstream applications such as drug sensitivity prediction and multi-omics data integration could deepen researchers' understanding of single-cell biology, which we leave as future work.

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A Sankey Plot

We provide the Sankey plot of single-cell foundation language models in Figure 3.

B Summary of Downstream Tasks

We provide a summary of single-cell data analysis tasks tailored for evaluating the existing foundation language models, which can be broadly categorized into common cell-level and gene-level tasks (Table 3), drug-related tasks (Table 4), spatial-related tasks (Table 4), and other relevant tasks (Table 4).

C Introduction of Benchmark Datasets

In this section, we list a variety of datasets for different single-cell foundation language models.

C.1 Datasets for Pre-training

As illustrated in Table 1, the existing single-cell PLMs opt for various datasets for pre-training. The introduction of these datasets is as follows:

- **scBERT** (Yang et al., 2022) leveraged around 1M scRNA-seq data from the Panglao dataset (Franzén et al., 2019) for pre-training.
- **UCE** (Rosen et al., 2023) generated the Integrated Mega-scale Atlas (IMA) dataset for pre-training, incorporating data from diverse sources, such as CELL×GENE (Abdulla et al., 2023) and Tabula Sapiens v1 (Consortium* et al., 2022). IMA encompasses a total of 36M cells.
- **GeneFormer** (Theodoris et al., 2023) generated Genecorpus-30M for pre-training, which consists of 27.4M transcriptomics data from publicly available sources.
- **CellPLM** (Wen et al., 2023) collected 9M scRNA-seq cells and 2M SRT cells from public data for pre-training.
- **scFoundation** (Hao et al., 2024) collected over 50M scRNA-seq cells from Gene Expression Omnibus (GEO), Single Cell Portal, Human Cell Atlas (HCA), and European Bioinformatics Institute (EMBL-EBI) for pre-training.
- **Nicheformer** (Schaar et al., 2024) was pre-trained on SpatialCorpus-110M, a large-scale dataset consisting of 57M cells. It was collected from CELL×GENE (Abdulla et al., 2023), GEO (Edgar et al., 2002; Barrett et al., 2012), sfaira (Fischer et al., 2021), and HCA.
- **tGPT** (Shen et al., 2023) collected a large-scale single-cell transcriptomics dataset with 22M cells for pre-training.
- **scGPT** (Cui et al., 2024) employed 33M cells from CELL×GENE (Abdulla et al., 2023) for pre-training.
- **CellLM** (Zhao et al., 2023) utilized around 2M cells from the Panglao dataset (Franzén et al., 2019) and the CancerSCEM dataset (Zeng et al., 2022) for pre-training.
- **LangCell** (Zhao et al., 2024) established scLibary, a dataset comprising 27.5M paired scRNA-seq data and textual descriptions, for pre-training. The dataset was sourced from CELL×GENE (Abdulla et al., 2023).
- **scCello** (Yuan et al., 2024) collected 22M scRNA-seq data from CELL×GENE (Abdulla et al., 2023) for pre-training.
- **scPRINT** (Kalfon et al., 2024) collected 50M cells from CELL×GENE (Abdulla et al., 2023) for pre-training.
- **scMulan** (Bian et al., 2024a) collected hECA-10M, a subset of Human Ensemble Cell Atlas (hECA) (Chen et al., 2022) for pre-training.
- **GeneCompass** (Yang et al., 2024) collected scCompass-126M, a dataset comprising 126M transcriptomic data from humans and mice for pre-training.
- **CellFM** (Zeng et al., 2024) collected around 100M cells from multiple databases for pre-training, including National Center for Biotechnology Information (NCBI)-GEO (Barrett et al., 2012), European Nucleotide Archive (ENA) (Cochrane et al., 2007), Genome Sequence Archive (GSA) (Chen et al., 2021), and ImmPort (Bhattacharya et al., 2018).

C.2 Datasets for Downstream Tasks

For downstream data analysis tasks, the introduction of dataset information is as follows:

- **scBERT** (Yang et al., 2022) employed various datasets for experiments. These datasets include the Zheng68k dataset (Zheng et al., 2017), pancreas datasets (Baron et al., 2016; Muraro et al., 2016; Segerstolpe et al., 2016; Xin et al., 2016), the MacParland dataset (MacParland et al., 2018),

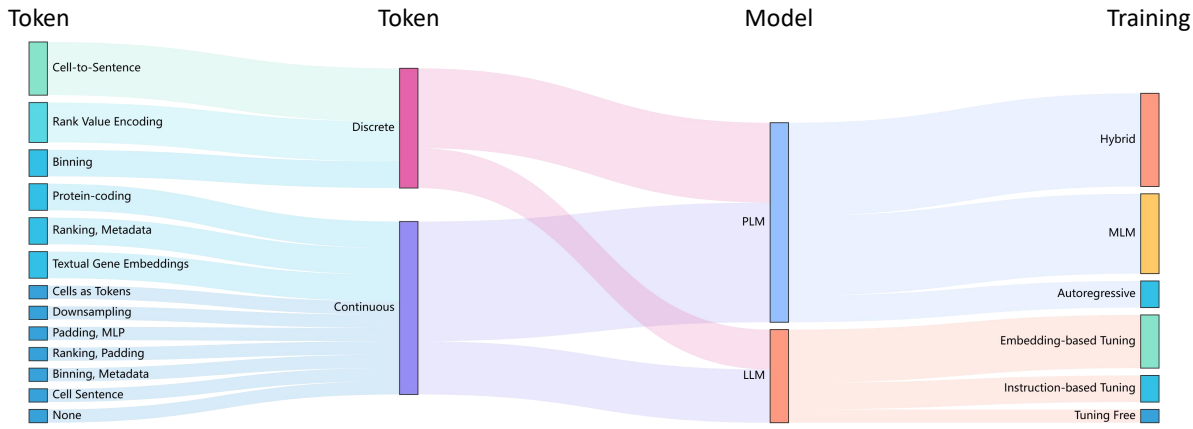


Figure 3: Sankey plot illustrating the flow of single-cell foundation language models, encompassing tokenization formats, model types, and training paradigms.

Model	Cell-level Tasks						Gene-level Tasks		
	Annotation	Discovery	Batch	Clustering	Integration	Generation	Network	Perturbation	Prediction
scBERT (Yang et al., 2022)	✓	✓							
UCE (Rosen et al., 2023)	✓	✓	✓	✓					
GeneFormer (Theodoris et al., 2023)	✓		✓	✓			✓	✓	
CellPLM (Wen et al., 2023)	✓			✓					
scFoundation (Hao et al., 2024)	✓			✓			✓	✓	
Nicheformer (Schaar et al., 2024)									
tGPT (Shen et al., 2023)			✓	✓					
scGPT (Cui et al., 2024)	✓		✓	✓	✓		✓	✓	
CellLM (Zhao et al., 2023)	✓								
scCello (Yuan et al., 2024)	✓	✓	✓	✓					✓
LangCell (Zhao et al., 2024)	✓	✓	✓	✓					
scPRINT (Kalfon et al., 2024)	✓		✓	✓			✓		
scMulan (Bian et al., 2024a)	✓		✓	✓		✓			
GeneCompass (Yang et al., 2024)	✓						✓	✓	✓
CellFM (Zeng et al., 2024)	✓						✓	✓	✓
Cell2Sentence (Levine et al., 2023)	✓					✓			
CHATCELL (Fang et al., 2024)	✓					✓			
scInterpreter (Li et al., 2024)	✓								
GenePT (Chen and Zou, 2024)	✓		✓	✓			✓		✓
scELMo (Liu et al., 2023a)	✓		✓	✓	✓		✓	✓	
CELLama (Choi et al., 2024)	✓								
scChat (Lu et al., 2024)	✓								

Table 3: Summary of common single-cell data analysis tasks.

heart datasets (Litviňuková et al., 2020; Tucker et al., 2020), the lung dataset (Lukassen et al., 2020), and the HCA dataset (He et al., 2020).

- **UCE** (Rosen et al., 2023) leveraged several benchmarks and datasets for evaluation, including single-cell integration benchmark (Luecken et al., 2022), cell ontology tree (Bard et al., 2005), Tabula Sapiens v2, Immune Cell Atlas (Domínguez Conde et al., 2022), green monkey lymph node and lung cells (Speranza et al., 2021), naked mole rat spleen and circulating immune cells (Hilton et al., 2019), chick retina (Yamagata et al., 2021), developing chick heart (Mantri et al., 2021), and mouse renal cells (Kragesteen et al., 2023).

- **GeneFormer** (Theodoris et al., 2023) was evaluated on various datasets, including fibroblasts (Xing et al., 2020), iPSC-derived myogenic cells (Guo et al., 2022), the aortic aneurysm dataset (Li et al., 2020), Drop-seq data (Selewa et al., 2020), dosage-related gene sets (Lek et al., 2016; Shihab et al., 2017; Ni et al., 2019), ESCs2 transcriptomes (Franzén et al., 2019), and Heart Atlas (Litviňuková et al., 2020).

- **CellPLM** (Wen et al., 2023) was evaluated on various datasets, including PBMC 5K and Jurkat from 10x Genomics, MERSCOPE FFPE Human Immuno-oncology data, lung cancer data (Li et al., 2020), liver cancer data (Ma et al., 2021), the Adamson Perturb-Seq dataset (Adamson et al., 2016), the Norman Perturb-Seq dataset (Norman et al., 2019), the hPancreas dataset (Chen et al., 2023), and the Multiple Sclerosis

Model	Drug-related Tasks	Spatial-related Tasks	Other Tasks
CellPLM (Wen et al., 2023)	-	Spatial Transcriptomic Imputation	scRNA-seq Denoising
scFoundation (Hao et al., 2024)	Drug Response Prediction	-	-
Nicheformer (Schaar et al., 2024)	-	Spatial Label Prediction, Spatial Composition	-
tGPT (Shen et al., 2023)	-	-	Inference of Developmental Lineages
CellLM (Zhao et al., 2023)	Drug Sensitivity Prediction	-	-
scCello (Yuan et al., 2024)	Drug Response Prediction	-	-
LangCell (Zhao et al., 2024)	-	-	Cell-text Retrieval, Pathway Identification
GeneCompass (Yang et al., 2024)	Drug Response Prediction	-	-
Cell2Sentence (Levine et al., 2023)	-	-	Cell Label Prediction
CHATCELL (Fang et al., 2024)	Drug Sensitivity Prediction	-	-
CELLama (Choi et al., 2024)	-	Spatial Context Analysis	-
scChat (Lu et al., 2024)	-	-	In-depth Analysis and Explanation

Table 4: Summary of drug-related, spatial-related, and other single-cell data analysis tasks.

(MS) dataset (Schirmer et al., 2019).

- **scFoundation** (Hao et al., 2024) used several datasets for downstream tasks. These datasets include the Baron dataset (Huang et al., 2018), the Zheng68K dataset (Zheng et al., 2017), the Segerstolpe dataset (Abdelaal et al., 2019), the CDR dataset (Liu et al., 2020), the drug response dataset (Zheng et al., 2023), the perturbation dataset (Roohani et al., 2022), and the cell mapping dataset (Cowan et al., 2020).

- **Nicheformer** (Schaar et al., 2024) was evaluated on the MERFISH mouse brain dataset (Yao et al., 2023), CosMx human liver and lung datasets (He et al., 2021), as well as Xenium human lung and colon datasets (from 10x Genomics).

- **tGPT** (Shen et al., 2023) was evaluated on the HCA dataset (Regev et al., 2018), the Human cell Landscape (HCL) dataset (Han et al., 2020), the Tabula Muris dataset (Schaum et al., 2018), the Cancer Genome Atlas (TCGA) dataset, and the Genotype-Tissue Expression Project (GTEx) dataset (Zhang et al., 2019).

- **scGPT** (Cui et al., 2024) leveraged several datasets for different tasks. For cell type annotation, it was evaluated on the MS dataset (Schirmer et al., 2019) and the myeloid dataset (Cheng et al., 2021). For other tasks, the experiments were conducted on the human pancreas dataset (Chen et al., 2023), the Lung-Kim dataset (Gavish et al., 2023), the COVID-19 dataset (Lotfollahi et al., 2021), the Norman and Adamson datasets, the Replogle dataset (Replogle et al., 2020), the PBMC 10k dataset, the perirhinal cortex dataset (Siletti et al., 2023), the 10x Multiome PBMC dataset, the BMMC dataset (Luecken et al., 2021), the ASAP PBMC dataset (Zhang et al., 2022), and the Immune Human dataset (Luecken et al., 2022).

- **CellLM** (Zhao et al., 2023) employed various datasets for different tasks. For cell type annotation, it leveraged the Zheng68k (Zheng et al., 2017) dataset and the Baron dataset (Baron et al., 2016). For drug sensitivity prediction, it was evaluated on human lung cancer cells (Aissa et al., 2021) and human oral squamous cancer cells (Sharma et al., 2018; Ravasio et al., 2020; Suphavilai et al., 2021).

- **LangCell** (Zhao et al., 2024) utilized a variety of benchmark datasets to evaluate the performance, including human peripheral blood cell datasets (Gayoso et al., 2022; Zheng et al., 2017), human liver datasets (Lin et al., 2020), the human brain cell dataset (Siletti et al., 2023), and the human cell dataset (Consortium* et al., 2022).

- **scCello** (Yuan et al., 2024) generated one in-distribution (ID) dataset and six out-of-distribution (OOD) datasets from CELL×GENE (Abdulla et al., 2023) for experiments.

- **scPRINT** (Kalfon et al., 2024) was evaluated on various datasets, encompassing kidney, retina, and colon tissues (Marshall et al., 2022; Wang et al., 2022; Kong et al., 2023), as well as ciliary body, colon, and retina tissues (van Zyl et al., 2022; Burclaff et al., 2022). Additionally, it was evaluated on human prostate tissues (Joseph et al., 2021), perturb-seq data (Dixit et al., 2016; Replogle et al., 2022), ChIP-seq data (Park, 2009), the pancreas dataset (Luecken et al., 2022), and the lung dataset (Sikkema et al., 2023).

- **scMulan** (Bian et al., 2024a) leveraged many datasets for evaluation, including the hECA-10M dataset, the heart cell dataset (Simonson et al., 2023), the liver dataset (Suo et al., 2022), the bone marrow dataset (He et al., 2020), the Human Cell Landscape dataset, the Lung integration dataset (Luecken et al., 2022), and the COVID-19 integration dataset (Lotfollahi et al., 2022).

- **GeneCompass** (Yang et al., 2024) employed various datasets for downstream tasks, including PBMC datasets (Qiao et al., 2013), human datasets (multiple sclerosis, lung, and liver), mouse datasets (brain, lung, and pancreas), the Immune Human dataset (Cui et al., 2024), and the drug dataset (Srivatsan et al., 2020).

- **CellFM** (Zeng et al., 2024) leveraged the Panglao dataset (Franzén et al., 2019) for gene function prediction, the Adamson and Norman datasets for perturbation prediction (Roohani et al., 2024), eight intra-datasets for cell annotation (Liu et al., 2023b), and gene datasets (Roohani et al., 2024; Tran et al., 2020; Luecken et al., 2022) for gene network analysis.

- **Cell2Sentence** (Levine et al., 2023) was evaluated on both single-cell data and bulk data, including the immune tissue data (Domínguez Conde et al., 2022), the cytokine stimulation dataset (Dong et al., 2023), the multi-tissue data (Megill et al., 2021), the Human PBMC data (Dong et al., 2023), the L1000 bulk RNA data (Subramanian et al., 2017), and the GTEx dataset (Consortium, 2020).

- **CHATCELL** (Fang et al., 2024) utilized the SHARE-seq mouse skin dataset (Ma et al., 2020) and the human lung cancer cells data (Aissa et al., 2021; Sharma et al., 2018; Ravasio et al., 2020; Suphavilai et al., 2021) for downstream tasks.

- **scInterpreter** (Li et al., 2024) constructed two datasets HUMAN-10k and MOUSE-13k for downstream tasks.

- **GenePT** (Chen and Zou, 2024) utilized various datasets for different tasks. For gene-level tasks, it was evaluated on a subset of datasets from GeneFormer (Theodoris et al., 2023) and Gene2vec (Du et al., 2019). For cell-level tasks, it was evaluated on a subset of datasets from scGPT (Cui et al., 2024), the Cardiomyocyte dataset (Chaffin et al., 2022), and the Aorta dataset (Li et al., 2020).

- **scELMo** (Liu et al., 2023a) was evaluated on CITE-seq data (Stoeckius et al., 2017), CyTOF data (Cheung and Utz, 2011), the PBMC dataset, the hPancreas dataset, the Aorta dataset, the Heart dataset, and perturb-seq-based datasets.

- **CELLama** (Choi et al., 2024) was evaluated on the Tabula Sapiens dataset (Consortium* et al., 2022), the COVID-19 dataset (Lotfollahi et al., 2021), human pancreas scRNA-seq data (Luecken et al., 2022), 10x genomics datasets, and human lung cell atals data (Travaglini et al., 2020).

- **scChat** (Lu et al., 2024) was evaluated on two scRNA-seq datasets from (Bagley et al., 2024) and (Mathewson et al., 2021).